THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE UNIT TEST SET C

Chapters 1–4: **Exploring the Investigative World of Science, The Invisible Living World: Beyond Our Naked Eye, Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. In a systematic investigation, a scientist begins with: 1

(a) a printed result **(b)** a final conclusion
(c) a focused question **(d)** an exam paper

2. Scientists studying whether the world is getting warmer rely on: 1

(a) rumours **(b)** luck
(c) observation, measurement and experiment **(d)** opinions only

3. When a scientist asks ‘What happens if I change this?’, she is: 1

(a) memorising **(b)** writing a poem
(c) starting an investigation **(d)** ending an investigation

4. Spirulina is a: 1

(a) fungus **(b)** microalga used as food supplement
(c) virus **(d)** protozoan

5. The basic unit of life is the: 1

(a) tissue **(b)** cell **(c)** organism **(d)** organ

6. The correct order of levels of organisation is: 1

(a) cell → organ → tissue → organism
(b) organ → cell → tissue → organism
(c) cell → tissue → organ → organ system → organism
(d) tissue → cell → organ → organism

7. Kamal Ranadive studied the link between viruses and: 1

(a) smallpox **(b)** malaria **(c)** cancer **(d)** diabetes

8. Which is a good way to prevent water-borne diseases? 1

(a) Drinking boiled water **(b)** Eating raw street food
(c) Drinking unfiltered pond water **(d)** Sharing glasses

9. Which disease is caused by a virus? 1

(a) Tuberculosis **(b)** Typhoid **(c)** Cholera **(d)** Dengue

10. The strength of an electromagnet increases when: 1

(a) more cells are connected **(b)** the iron core is removed
(c) fewer cells are used **(d)** the circuit is opened

11. Why should we not touch a nichrome wire for long when current passes? 1

(a) it becomes soft **(b)** it becomes hot
(c) it becomes cold **(d)** it becomes magnetic

12. The magnetic effect of electric current was discovered by: 1

(a) Hans Christian Oersted **(b)** Luigi Galvani
(c) Michael Faraday **(d)** Alessandro Volta

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 10–15 words each.

13. Name two instruments you would use in a puri-frying investigation and what each measures. 2

14. Give two examples of unicellular organisms. 2

15. Distinguish between a symptom and a sign with one example each. 2

16. Name two appliances that use heating element. 2

17. State one use of a lifting electromagnet. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 30–40 words each.

18. Why does the chapter say 'science is everywhere'? 3

19. Write a short note on viruses. 3

20. Why are NCDs increasing in India? Give three reasons. 3

21. Describe the lemon cell activity. 3

22. Why can microorganisms be found almost everywhere? 3

OR

Explain how legume roots and Rhizobium help each other. 3

23. Explain how antibiotics work and why they fail against viruses. 3

OR

Describe how antibiotic resistance develops. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

24. Describe the yeast activity with two test tubes A and B. Write the set-up, observation and conclusion. 5

OR

Compare plant, animal and bacterial cells in a table of at least five points. 5

25. Discuss the story of Galvani and Volta and its importance in the development of the battery. 5

OR

Explain Oersted's observation and show how an electromagnet is made. Name three factors that affect its strength. 5

END OF QUESTION PAPER — Total: 50 Marks

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